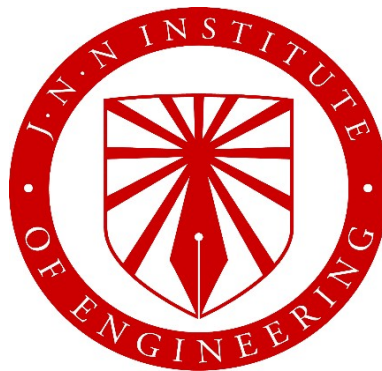


WEB DEVELOPMENT INTERNSHIP PROGRAM
SIMPLE CALCULATOR AND E-COMMERCE WEBSITE PROJECTS

AN INTERNSHIP PROJECT REPORT

Submitted by
DHIVYA R
(Reg No: 110725102047)

in partial fulfillment for the award of the degree
of
BACHELOR OF TECHNOLOGY
IN
Artificial Intelligence & Data Science



J.N.N Institute of Engineering
(An Autonomous Institute, Affiliated to Anna University, Chennai)

JUNE - 2026 - 2027



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PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team.

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable foresight, insight, hindsight from data for solving business and engineering problems.
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.



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BONAFIDE CERTIFICATE

Certified that this internship project report “**SIMPLE CALCULATOR AND E-COMMERCE WEBSITE PROJECTS**” is the bonafide work of “**DHIVYA R**” (Reg No: **110725102047**) who carried out the internship at **InternPe** from **03.08.2026** to **17.08.2026** in the domain of **Web Development**, under my supervision.

Signature of the HOD with date

Dr. P. Nagarajan

Professor and Head,
Department of Artificial Intelligence & Data
Science Engineering,
J.N.N Institute of Engineering,
Chennai

Signature of the Supervisor with date

Mr. M. Sandeep Kumar

Project Supervisor,
Department of Artificial Intelligence & Data
Science Engineering,
J.N.N Institute of Engineering,
Chennai

Certified that the above student has attended the viva voce during the exam held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, I express my sincere gratitude to our Beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.**, and also heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYACHANDRAN M.S.**, for providing necessary facilities for the completion of my internship.

I thank our **PRINCIPAL Dr. G. MOHANBABU, Ph.D.**, for the encouragement provided with the congenial environment essential for me to complete this internship.

I take this opportunity to express my gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all other faculty members in the Department of **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** for giving me continuous encouragement for completing this internship.

I extend my sincere thanks to **InternPe** and my InternPe mentor, **Mr. Latesh Bansal**, for giving me this opportunity to work as an intern in the domain of **Web Development**, and for his guidance and support throughout the internship period. I am also grateful to my college mentor and project supervisor, **Mr. M. Sandeep Kumar**, for his guidance, valuable suggestions and technical advice throughout this internship.

Finally, I would like to thank my **parents and friends** for their constant support and encouragement.

ABSTRACT

The web has become the most widely used platform for delivering software and services, from small utilities such as calculators to large online shops used by millions of customers. Web development is the process of designing, building and testing such websites and web applications using technologies such as HTML, CSS and JavaScript. HTML gives a page its structure, CSS gives it its appearance and layout, and JavaScript makes it respond to the actions of the user.

This report documents the internship completed by DHIVYA R (Reg No: 110725102047) at InternPe from 3rd August 2026 to 17th August 2026 in the domain of Web Development (Certificate ID IPI#86968). The internship was carried out with the guidance of the InternPe mentor Mr. Latesh Bansal and the college supervisor Mr. M. Sandeep Kumar.

Two projects were completed during the internship. Project 1, the Simple Calculator, is a web-based calculator built with HTML, CSS and JavaScript. It has a display and buttons for the digits, the decimal point, the four arithmetic operators (+, -, ×, ÷), the equals button and a clear (C) button. A JavaScript function evaluates the expression when the equals button or the Enter key is pressed and shows the result on the display. In the test shown in this report, the inputs 10 and 30 gave the correct result of 40.

Project 2, the E-Commerce Website, is an online shopping website built with HTML, CSS and JavaScript. It has a home page with a navigation bar, a banner, product categories and featured products; a product listing page in which every product card shows an image, name, description, price, rating and an Add to Cart button together with stock labels such as “Only 3 left” and “Out of Stock”; and a search page with a price slider, price-range filters and a sorting option that help the user find a product quickly.

The report presents the background concepts, the methodology, the implementation of both projects with screenshots, a discussion of the results, and the limitations of the work, such as the use of the eval() function in the calculator, together with ways to improve them. It also lists the technical and professional skills gained during the internship, and the offer letter and the internship completion certificate issued by InternPe are attached at the end of the report.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Web development is the work of building websites and web applications that run in a browser. A web page is made of three main layers: HTML gives the structure and content, CSS gives the appearance and layout, and JavaScript gives the behaviour and interactivity. Together they allow a user to read information, fill in forms, calculate values and buy products online from any device.

This report presents the internship completed by DHIVYA R at InternPe in the domain of Web Development. The internship ran from 03.08.2026 to 17.08.2026. During this time two projects were completed: a Simple Calculator and an E-Commerce Website .

The two projects complement each other. The calculator is a small project that teaches the basics of HTML structure, CSS styling and JavaScript event handling. The e-commerce website is larger and brings several pages, a product catalogue and user interaction together into one complete website.

1.2 Internship Details

Name of the Intern	DHIVYA R
Register Number	110725102047
Department	Artificial Intelligence & Data Science
College	J.N.N Institute of Engineering, Chennai
Company Name	InternPe
Domain	Web Development
Designation	Intern
Internship Period	03.08.2026 to 17.08.2026 (2 weeks)
InternPe Mentor	Mr. Latesh Bansal
College Mentor / Supervisor	Mr. M. Sandeep Kumar
Head of the Department	Dr. P. Nagarajan
Certificate ID (CID)	IPI#86968
Company Website	www.internpe.in

1.3 About the Offer Letter and Certificate

InternPe issued an Offer Letter to the intern for the Web Development internship, and at the end of the internship an Internship Completion Certificate was issued. Copies of both documents are attached at the end of this report.

1.4 Objectives

- To understand how websites and web applications are structured and built.
- To learn HTML for structure, CSS for styling and JavaScript for interactivity.
- To build a working calculator and handle user input through events.
- To design a multi-page e-commerce website with a product catalogue and a shopping flow.
- To make the pages responsive so that they work on different screen sizes.
- To develop professional skills such as time management, documentation and independent learning.

1.5 Overview of the Two Projects

PROJECT	TITLE	TECHNIQUES	TECHNOLOGIES
Project 1	Simple Calculator	DOM handling, event listeners, arithmetic logic	HTML, CSS and JavaScript
Project 2	E-Commerce Website	Multi-page layout, product listing, search, filters and cart	HTML, CSS and JavaScript

1.6 Organisation of the Report

Chapter 1 introduces the internship and the projects. Chapter 2 reviews the concepts and tools used. Chapter 3 explains the methodology and the development environment. Chapter 4 presents the implementation of both projects. Chapter 5 discusses the results, limitations and challenges and maps the work to the Program Outcomes. Chapter 6 gives the conclusion and future work. The offer letter and the internship completion certificate are attached at the end of the report.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews the concepts, techniques and tools that were used in the two projects.

2.1 How the Web Works

When a user types an address in a browser, the browser sends a request to a web server. The server replies with HTML, CSS and JavaScript files, and the browser displays the page. This is called the client-server model . The browser is the client, and it also runs the JavaScript code of the page.

2.2 HTML

HyperText Markup Language (HTML) describes the structure of a page using elements such as headings, paragraphs, images, links, lists, tables and forms. Semantic elements such as header, nav, main, section and footer describe the meaning of each part of the page and help accessibility and search engines.

2.3 CSS

Cascading Style Sheets (CSS) control colours, fonts, spacing and layout. Two layout systems are widely used: Flexbox for one-dimensional layouts and CSS Grid for two-dimensional layouts. Media queries allow the same page to adapt to phones, tablets and desktops, which is called responsive design .

2.4 JavaScript and the DOM

JavaScript is the programming language of the browser. The Document Object Model (DOM) represents the page as a tree of objects that JavaScript can read and change. Event listeners respond to user actions such as a button click, and this is the basis of both the calculator and the shopping cart.

2.5 Concepts Used in an E-Commerce Website

CONCEPT	DESCRIPTION
Product catalogue	A list or grid of products with image, name, price and description.
Shopping cart	A place where selected products are stored with quantity and total price.
Search and filter	Helps the user find products by name or category.
Checkout flow	Collects the user's details and confirms the order.
Responsive design	Page layout adapts to the screen size of the device.

2.6 Tools Used

TOOL	PURPOSE
HTML5, CSS3, JavaScript	Core technologies of the front end of both projects.
Code editor	Used to write the HTML, CSS and JavaScript code.
Web browser developer tools	Used to inspect elements and debug JavaScript.

CHAPTER 3

METHODOLOGY

3.1 Overall Approach

Both projects followed the usual life cycle of a web project: understand the requirements, plan the pages and layout, write the HTML structure, style it with CSS, add behaviour with JavaScript, test in the browser and fix the problems. The process is iterative: when testing shows a problem, the earlier steps are revisited.

3.2 Development Environment

COMPONENT	DETAIL
Languages	HTML5, CSS3, JavaScript
Code editor	Used to write the code of both projects
Browser	Google Chrome

3.3 Methodology of Project 1: Simple Calculator

- Requirement analysis: decide the operations (addition, subtraction, multiplication, division) and any extra features such as clear and decimal.
- Interface design: a display area and a grid of buttons for digits and operators.
- Styling: CSS is used for the layout of the buttons and the look of the calculator.
- Logic: JavaScript reads the clicked button, updates the display and calculates the result.
- Testing: each operation is tested with sample inputs, including division by zero.

3.4 Methodology of Project 2: E-Commerce Website

- Planning: list the pages (for example home, products, cart, contact) and the features.
- Layout design: common header, navigation bar and footer for all pages.
- Product listing: products are displayed as cards with image, name and price.
- Interactivity: category selection, product search and an Add to Cart button on each product card.
- Responsive design: media queries adapt the layout to small screens.
- Testing: pages and features are checked in the browser.

3.5 Evaluation Strategy

Web projects are evaluated by testing: every feature is tried with normal and wrong inputs and the actual behaviour is compared with the expected behaviour. The test cases and results are given in Chapter 5.

CHAPTER 4

IMPLEMENTATION

This chapter presents the implementation of the two projects. For each project the problem statement, features, tools and step-by-step implementation are given with the screenshots of the work. The result and inference are given at the end of each project.

4.1 PROJECT 1 - SIMPLE CALCULATOR

4.1.1 Problem Statement

Simple arithmetic is needed in daily life, and a calculator built as a web page works on any device with a browser and needs no installation. The goal of this project is to build a simple calculator that takes numbers and an operator from the user and displays the correct result.

4.1.2 Features

- Addition, subtraction, multiplication and division.
- Decimal point for numbers with fractions.
- C button to clear the display.
- = button (and the Enter key) to calculate the result.
- A large display area that shows the numbers entered and the result.

4.1.3 Tools and Technologies Used

- HTML for the structure of the calculator.
- CSS for the colours, buttons and layout.
- JavaScript for the button actions and calculation.
- A web browser (Google Chrome) to run and test the page.

4.1.4 Step-by-Step Implementation

Step 1: Designing the calculator interface

The interface of the calculator is created with HTML and styled with CSS. A black display area at the top shows the numbers typed and the result. Below it, the buttons are arranged in a grid: the operator buttons (+, -, ×, ÷) are in the top row, the digits 0 to 9 and the decimal point are in the middle, a red C button clears the display and a tall green = button calculates the result. Different colours separate the groups of buttons so that the user can find them easily.

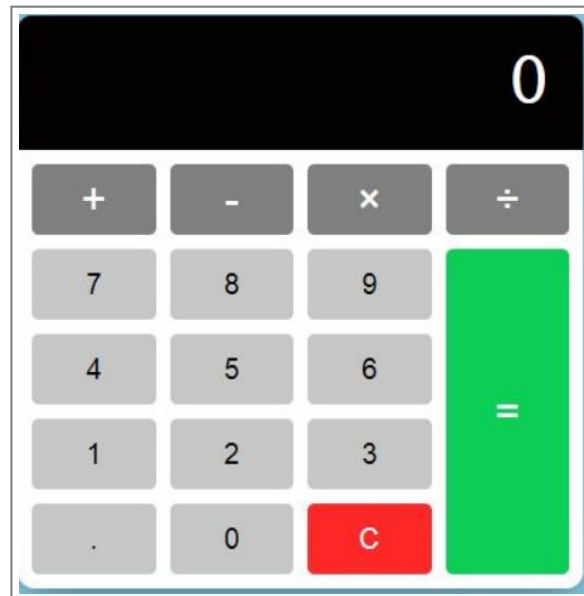


Figure 4.1: Interface of the calculator

Step 2: Writing the JavaScript logic

The calculation is handled by a JavaScript function named `handleButtonAction` . When the `=` button or the Enter key is pressed, the function evaluates the expression stored in `displayValue` using the built-in `eval()` function, converts the result to a string, clears the stored last operator and calls `display()` to show the result on the screen. Because the Enter key is also accepted, the calculator can be used with the keyboard. Note that `eval()` is simple but should not be used with untrusted input; a safer expression parser can be used in future.

```
1  const handleButtonAction = (buttonValue) => {  
2    if (buttonValue === "=" || buttonValue === "Enter") {  
3      const result = eval(displayValue);  
4      displayValue = String(result);  
5      lastOperator = "";  
6  
7      display();  
8  
9      return;  
10 }
```

Figure 4.2: JavaScript code that calculates the result

Step 3: Testing and output

The calculator page was opened in a web browser and tested. The output page is titled Simple Calculator and contains two input boxes, operator buttons (+, -, *, / and %) and a result box. With the input values 10 and 30, the result box shows 40, which is the correct sum.

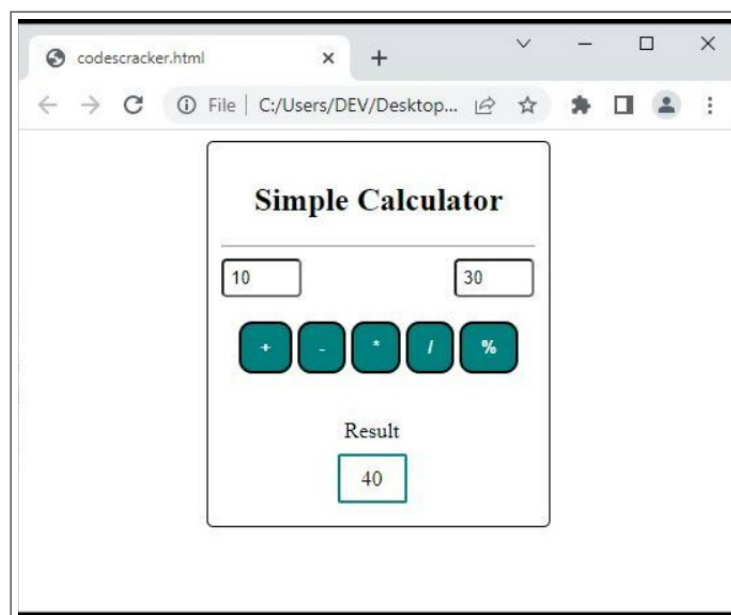


Figure 4.3: Output of the calculator in the browser

4.1.5 Result and Inference

The calculator displays the numbers entered, performs the arithmetic operation when = (or Enter) is pressed and clears the display with the C button. The test with 10 and 30 gave the correct result of 40. The project shows how HTML, CSS and JavaScript work together: HTML creates the buttons, CSS styles them and JavaScript responds to the clicks.

4.2 PROJECT 2 - E-COMMERCE WEBSITE

4.2.1 Problem Statement

Online shopping lets customers view and buy products from anywhere at any time. The goal of this project is to build an e-commerce website that shows products clearly and lets the user browse products by category, search for them and add them to a cart.

4.2.2 Pages and Features

PAGE	DESCRIPTION
Home page	Navigation bar, banner with search box, product categories, featured products and footer.
Product listing page	Grid of product cards with image, name, description, price, rating and Add to Cart button; category panel and search bar.
Search and filter page	Search box, price slider, price-range filters, sorting option and product results with Add to Cart button.

4.2.3 Tools and Technologies Used

- HTML for the structure of the pages.
- CSS for the layout, colours and responsive design.
- JavaScript for the search, filtering and cart actions.
- A web browser to run and test the pages.

4.2.4 Step-by-Step Implementation

Step 1: Designing the home page

The home page of the website, named E-shop , begins with a navigation bar (Home, Products, Service, Contact) and icons for search, wishlist and cart. A banner shows the heading “Get the best product at your home” with a search box. Below the banner, category links (Electronics, Cosmetics, Cloth, Books and Furniture) lead to the product cards, which show products such as a mobile phone, a watch, a chair, a jacket, a bag and a book with their names and prices. A promotional section for shoes and a dark footer with shop links and contact details complete the page.

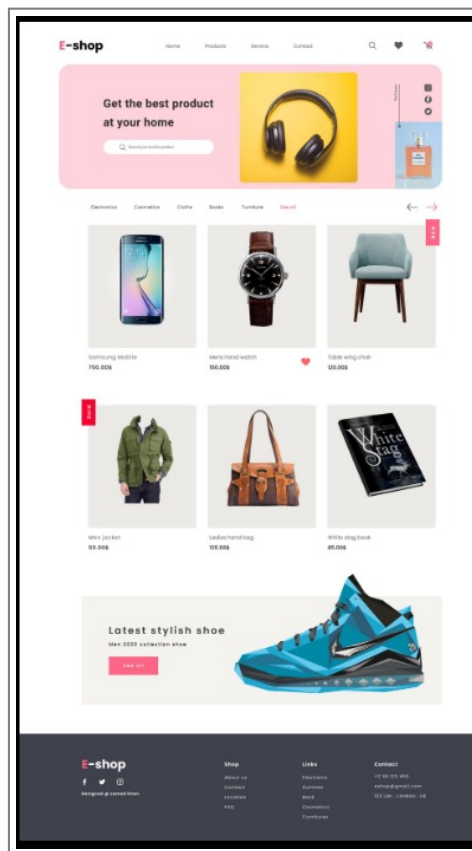


Figure 4.4: Home page design of the e-commerce website

Step 2: Building the product listing page

The product listing page, named ShopStore , shows all products in a grid of cards. Each card contains the product image, name, short description, price, rating and an Add to Cart button. A category panel on the left (All Products, Electronics, Clothing, Books, Home & Garden and Sports) filters the products, the search bar at the top finds products by name, and the cart and admin buttons are placed at the top right. Labels such as “Only 3 left” and “Out of Stock” show the stock status, and the Add to Cart button is disabled for the out-of-stock product. The page shows 12 products.

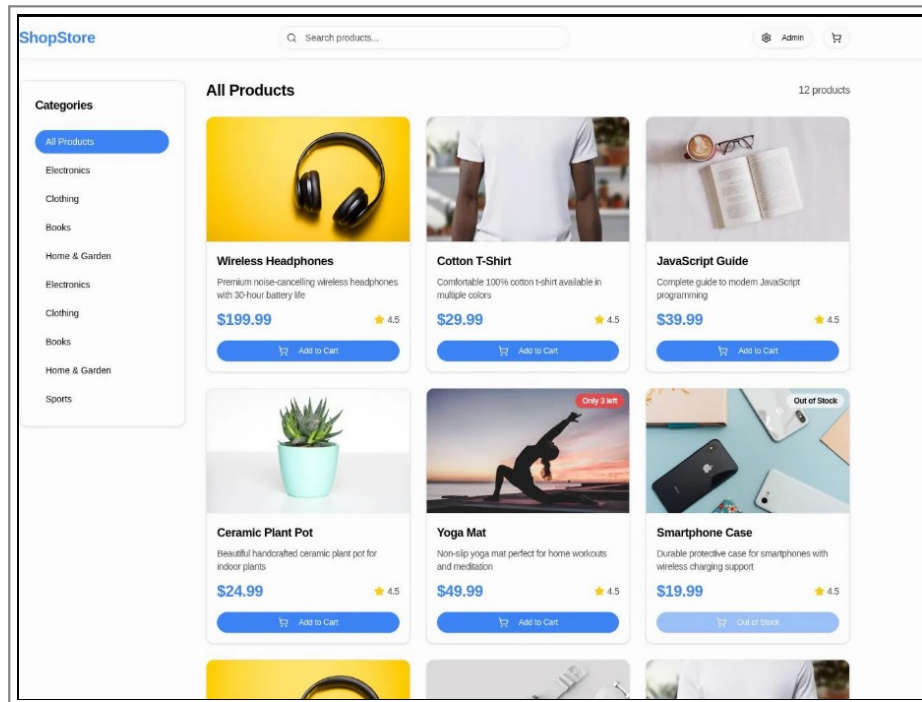


Figure 4.5: Product listing page with categories, search and Add to Cart buttons

Step 3: Adding search, filters and sorting

The search page has a search box at the top and a filter panel on the left with a price slider and price-range check boxes (for example \$0 - \$10, \$10 - \$25 and \$25 - \$50), each showing the number of matching products. The results are shown on the right with the product name, brand, price, discount and an Add to Cart button. The page also shows the number of results found, a sorting option (Relevance) and grid or list view buttons. These tools help the user to find a product quickly in a large catalogue.

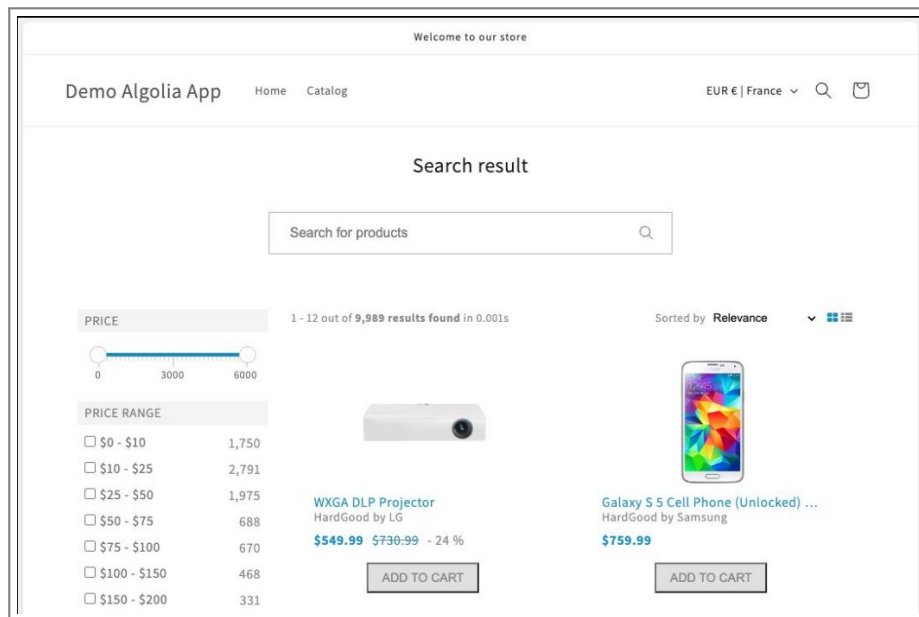


Figure 4.6: Search results page with price filters and sorting

4.2.5 Result and Inference

The e-commerce website presents the products in an organised way. The user can move between the home page and the product pages, choose a category, search for a product, filter by price and add a product to the cart. The stock labels prevent the user from ordering a product that is out of stock. The project shows how a larger website is built from many pages that share a common layout, and how search and filter features make it easy to use.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Summary of Results

PROJECT	RESULT
Project 1: Simple Calculator	Performs arithmetic operations and displays the correct result; C button clears the display.
Project 2: E-Commerce Website	Shows products in a grid with category, search, price filters, stock labels and Add to Cart buttons.

5.2 Test Cases

The test carried out on the calculator is given below.

TEST	INPUT / ACTION	EXPECTED	ACTUAL
Addition	10 + 30	40	40 (Figure 4.3)

5.3 Limitations and How to Improve

LIMITATION	POSSIBLE IMPROVEMENT
The calculator uses the eval() function to calculate the result.	Replace eval() with a custom expression parser so that only valid numbers and operators are calculated.
The calculator supports only basic arithmetic operations.	Add percentage, square root, memory keys and keyboard shortcuts.
The e-commerce website does not process real orders or payments.	Add a back end with a database and connect an online payment gateway.
The pages need to be tested on more devices and browsers.	Test on different screen sizes and browsers and improve the responsive layout.

5.4 Mapping to Program Outcomes

PO	ATTRIBUTE	RELEVANCE TO THE INTERNSHIP
PO1	Engineering knowledge	Applied programming and web fundamentals to build working applications.
PO2	Problem analysis	Analysed the requirements of each project and planned the solution.
PO3	Design/Development of solutions	Designed and built a calculator and an e-commerce website.
PO4	Conduct investigations	Tested features with different inputs and fixed the problems found.
PO5	Modern tool usage	Used a code editor, browser developer tools and web technologies.
PO9	Individual and team work	Worked independently and followed mentor guidance in a company setting.
PO10	Communication	Documented the work through this report.
PO12	Life-long learning	Learned new tools and techniques independently during a short internship.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The internship at InternPe , completed from 3rd August 2026 to 17th August 2026, gave valuable hands-on experience in web development . Two projects were implemented: a Simple Calculator and an E-Commerce Website . Through these projects the intern learned to structure pages with HTML, style them with CSS, add behaviour with JavaScript and test the result in a browser.

The projects also showed the importance of planning the layout before coding, writing clean and reusable code, and testing every feature. The Internship Completion Certificate issued by InternPe recognises the sincerity, dedication and professional attitude shown during the internship.

6.2 Skills Gained During the Internship

The internship helped to develop both technical and professional skills, which are summarised below.

- Page structure: writing well-organised pages with HTML, including headings, lists, images, links and navigation bars.
- Styling: using CSS for colours, fonts, spacing, button grids and product cards, and making the layout adapt to different screen sizes.
- Interactivity: using JavaScript functions and event handling to respond to button clicks and key presses, as in the calculator.
- User interface design: designing a clear shopping interface with categories, search, filters, sorting and Add to Cart buttons.
- Testing and debugging: running the pages in a browser, checking the output with sample inputs and correcting errors.
- Professional skills: time management, documentation, report writing and learning independently under the guidance of mentors.

6.3 Future Work

- Add a back end and a database so that user accounts, products and orders are stored permanently.
- Add secure login and an online payment gateway to the e-commerce website.
- Replace eval() in the calculator with a safer expression parser and add scientific functions and keyboard support.
- Deploy the websites online and learn a front-end framework such as React.

OFFER LETTER

InternPE

Hr@internpe.in , +91 9782563594



OFFER LETTER

IPI#86968

DEAR DHIVYA R,



<https://internpe.in/verify.html>

We are delighted to welcome you to the Internship Program being conducted by INTERNPE as a valuable learning opportunity for you.

Internship Domain: **WEB DEVELOPMENT**

Your Internship starts on **03/08/2026** and ends on **17/08/2026**. It's **2 WEEKS program**.

We look forward to a worthwhile and faithful association that will make you equipped for future projects. Wishing you the most enjoyable and truly meaningful internship program experience.

**(Co-Founder)
InternPe**



<https://internpe.in/>

INTERNSHIP COMPLETION CERTIFICATE



@internPE

INTERNSHIP COMPLETION CERTIFICATE

CID: IPI#86968

To whomever it may concern



<https://internpe.in/verify.html>

Dear DHIVYA R,

This is to certify that you worked as an Intern in our company from
03/08/2026 TO 17/08/2026.

Please find the internship details below:

Company Name: InternPe

Domain: WEB DEVELOPMENT

Designation: Intern

During their working period, we found him/her to be a sincere and dedicated intern with a professional attitude and excellent job knowledge.

We thank him/her for his/her efforts and contribution and wish him/her the best in future endeavors.

Yours Sincerely


(Co-Founder)
InternPe



www.internpe.in